Roadmap: You Are Here

Time	Part	Status
1:00–2:15	Part 4: Going Distributed	✓
2:30–3:15	**Part 5: Economics & Sustainability**	← **You are here**
3:15–3:45	Part 6: Design Space Exploration	
3:45–4:15	Part 7: TinyML to Frontier	
4:15–4:45	Part 8: Advanced Topics	
4:45–5:00	Part 9: Wrap-Up & Capstone	

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Economics & Sustainability

1

Key Question

How much does it *really* cost to train and serve an LLM—in dollars, kilowatt-hours, and tonnes of CO_2?

"A 1024-GPU H100 cluster costs $30M in hardware. Electricity is surprisingly small (∼10% of total). The dominant cost lever is utilization—idle GPUs cost the same as busy ones."

Wall 17: The Capital Wall (TCO)

The Equation

$$\text{TCO} = \underbrace{\text{CapEx}}_{\text{hw + facility}} + \underbrace{\text{OpEx}_{\text{energy}}}_{\text{electricity}} + \underbrace{\text{OpEx}_{\text{maint}}}_{\text{staff}}$$

Key insight: GPUs are only ∼40% of total CapEx. Networking, cooling, facility, and staff add 50–150%.

```
econ = mlsysim.
    EconomicsModel()
r = econ.solve(
  fleet=mlsysim.Systems
    .Clusters.Research_256,
  duration_days=30,
  infrastructure_multiplier
    =2.5)
print(f"TCO: ${r.tco_usd:,.0
    f}")
```

- `infrastructure_multiplier = 2.0–2.5` for full datacenter TCO
- Amortized over 3–5 year depreciation schedule

The Infrastructure Multiplier

Component	Cost (%)	Multiplier
GPU Hardware	40%	1.0×
Network (IB)	20%	—
Power & Cooling	15%	—
Facility / Land	10%	—
Staff / Operations	15%	—
Total	100%	**2.5**×

Pitfall: Budgeting only for GPUs underestimates TCO by 2–3×.

Wall 18: The Sustainability Wall

The Equation

$$CO_2 = \text{Energy} \times \text{PUE} \times \text{CI}$$

Three levers:

1. **Energy** — Better MFU, smaller model
2. **PUE** — Liquid cooling (1.3 → 1.06)
3. **Carbon intensity** — Choose your grid

```
sus = mlsysim.
    SustainabilityModel()
r = sus.solve(
  fleet=mlsysim.Systems
    .Clusters.Research_256,
  datacenter=mlsysim.Infra
    .Grids.Quebec,
  duration_days=30, mfu
   =0.45)
print(f"{r.
    carbon_footprint_kg
         /1000:.1f} tonnes
    CO2")
```

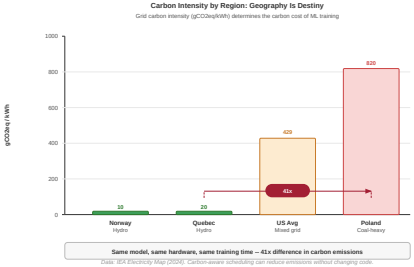
Carbon Intensity: The 41× Gap

Region	Mix	gCO_2
Quebec	Hydro	20
Sweden	Hydro+Nuc	25
US Avg	Mixed	390
Germany	Coal+Wind	380
Poland	Coal	820

Training the *same model*:

Poland: ∼800 t vs **Quebec:** ∼20 t

⇒ **41**× difference.



Embodied Carbon: Manufacturing Dominates at the Edge

Cloud / Training

- Operational carbon dominates
- GPU runs 24/7 for months
- Embodied: $\sim$5% of lifecycle

IoT / TinyML

- Manufacturing carbon dominates
- MCU uses microwatts
- Embodied: $>$90% of lifecycle
- Multiplied by millions of devices

Source: Gupta et al. (2022), “ACT: Designing Sustainable Computer Systems with an Architectural Carbon Modeling Tool”

Live Demo: TCO + Carbon Analysis

```
fleet = mlsysim.Systems.Clusters.Research_256
econ  = mlsysim.EconomicsModel()
result = econ.solve(
    fleet=fleet, duration_days=30,
    datacenter=mlsysim.Infra.Grids.Quebec,
    mfu=0.45, infrastructure_multiplier=2.5,
    amortization_years=3.0)
print(f"TCO: ${result.tco_usd:,.0f}")
print(f"Carbon: {result.carbon_footprint_kg/1000:.1f} t")
```

Live Demo: The Carbon Geography Experiment

```
fleet  = mlsysim.Systems.Clusters.Research_256
solver = mlsysim.SustainabilityModel()
for region in [mlsysim.Infra.Grids.Poland,
               mlsysim.Infra.Grids.Quebec]:
    r = solver.solve(fleet=fleet, duration_days=30,
                     datacenter=region, mfu=0.45)
    print(f"{r.region_name:>12}: "
          f"{r.carbon_footprint_kg/1000:.1f} t CO2")
```

Expected Output

```
  Poland:  1,064 MWh, 872.5 tonnes CO2
  Quebec:  1,064 MWh,  21.3 tonnes CO2
```

Same energy, 41× less carbon. Geography wins.

Exercise 5a: Carbon-Aware Placement

Your Task (3 minutes)

Compare a 30-day training run on 512 H100s: **Poland** vs **Quebec**. How many tonnes of CO_2 saved?

```
fleet = mlsysim.Fleet(
    name="Training Fleet",
    node=mlsysim.Systems.Nodes.DGX_H100,
    count=64,
    fabric=mlsysim.Systems.Fabrics.InfiniBand_NDR)
solver = mlsysim.SustainabilityModel()
# YOUR CODE: solve for Poland and Quebec
```

Exercise 5b: Multi-Vendor TCO Shootout

Your Task (5 minutes)

Same workload: Llama-3 70B, 30-day training run, 512 accelerators.

Which cluster is cheapest for the **same throughput**?

```
econ = mlsysim.EconomicsModel()
for cluster_name in ["H100", "MI300X", "Gaudi3"]:
    hw = getattr(mlsysim.Hardware.Cloud, cluster_name)
    fleet = mlsysim.Fleet(name=cluster_name, hw=hw,
                          count=64)
    r = econ.solve(fleet=fleet, duration_days=30,
                   infrastructure_multiplier=2.5)
    print(f"{cluster_name:>8}: TCO=${r.tco_usd:,.0f}")
```

The cheapest accelerator depends on the workload's binding constraint.

Key Takeaway: Economics & Sustainability

1. **TCO $\neq$ GPU cost.** Infrastructure is 2–2.5$\times$.
2. **Utilization is the cost lever.** Idle GPUs cost the same.
3. **Geography $>$ algorithms** for carbon reduction.

"The cheapest watt is the one you don't consume.
The cleanest watt is the one from a hydro dam."

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Design Space Exploration

2

Key Question

Given a budget and an SLA,

how do you find the *best* hardware configuration?

The search space is combinatorial:
$|\text{hardware}| \times |\text{batch sizes}| \times |\text{precisions}| \times |\text{parallelism configs}|$
can easily exceed 10^4 configurations.

The DSE Pattern: Declare, Search, Rank

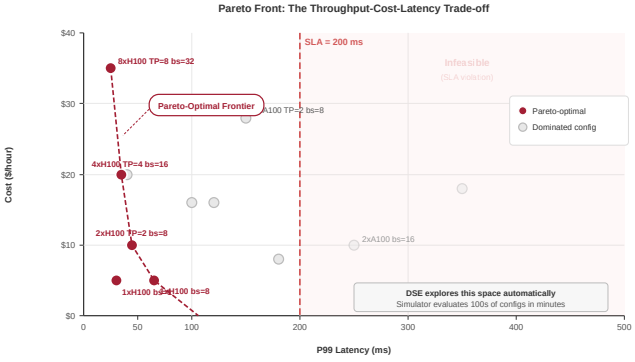
1. **Declare** the search space:
   - Hardware: {A100, H100, H200, B200}
   - Batch sizes: {1, 2, 4, . . . , 64}
   - Precisions: {fp16, int8, int4}
2. **Search** with an objective:
   - `minimize: tco_usd`
   - `maximize: throughput`
3. **Rank** subject to constraints:
   - `latency < 50 ms`
   - `feasible == True`

```
from mlsysim.core.dse import DSE

dse = DSE(
  space={
    "batch_size": [1,4,8,16,32],
    "precision": ["fp16","int8"
   ],
  },
  objective="maximize:
   throughput",
  constraints=["latency < 100"])
```

Analytical models ⇒ each evaluation takes <1 ms ⇒ exhaustive search.

Pareto Fronts: No Free Lunch



Each point is one hardware + parallelism + batch size configuration. Pareto front identifies non-dominated trade-offs.

Pareto front: the set of configurations where

Live Demo: Design Space Sweep

```
hw_list = [mlsysim.Hardware.Cloud.H100,
           mlsysim.Hardware.Cloud.MI300X,
           mlsysim.Hardware.Cloud.Gaudi3,
           mlsysim.Hardware.Cloud.B200]
results = mlsysim.Engine.sweep(
    llama, hw_list, batch_sizes=[1, 4, 16, 64],
    precisions=["fp16", "int8"], efficiency=0.5)
for r in sorted(results,
        key=lambda x: x['profile'].latency.magnitude):
    p = r['profile']
    print(f"{r['hardware']:>8} bs={r['batch_size']:<3} "
          f"| {p.bottleneck:<8} | MFU={p.mfu:.3f}")
```

Live Demo: DSE with Objective & Constraints

```
from mlsysim.core.dse import DSE

dse = DSE(
    space={"batch_size": [1,4,8,16,32,64],
           "precision": ["fp16", "int8"]},
    objective="maximize: throughput",
    constraints=["latency < 100"])
def evaluate(params):
    p = mlsysim.Engine.solve(llama, hw,
        batch_size=params["batch_size"],
        precision=params["precision"])
    return {"throughput": p.throughput.magnitude,
            "latency": p.latency.to("ms").magnitude}
best = dse.search(evaluate)
```

Live Demo: Batching Optimizer (Pareto Front)

```
opt = mlsysim.BatchingOptimizer()
result = opt.solve(
    model=llama, hardware=hw, seq_len=128,
    arrival_rate_qps=10.0,
    sla_latency_ms=20000.0, precision="fp16")
print(f"Optimal bs: {result.best_batch_size}")
for pt in result.pareto_front:
    print(f"  bs={pt['batch_size']:<4} "
          f"lat={pt['p99_latency'].m_as('ms'):.0f} ms")
```

Exercise: Budget-Constrained Design

Your Task (5 minutes)

Given a **$1M budget**, find the best hardware + batch size + precision configuration for serving Llama-3-8B inference under a 50 ms latency SLA.

```
# Hint: Use Engine.sweep across hardware tiers,
# filter by feasibility and latency SLA,
# then pick the highest-throughput config
# within budget (check unit_cost in Hardware)
```

Key Takeaway: Design Space Exploration

1. **Analytical models** make exhaustive DSE feasible ($<$1 s for 10^4 configs)
2. **Pareto fronts** reveal the actual tradeoff structure
3. **Constraints first**: filter infeasible, then optimize

"You cannot optimize what you cannot model.
You cannot model what you cannot measure."

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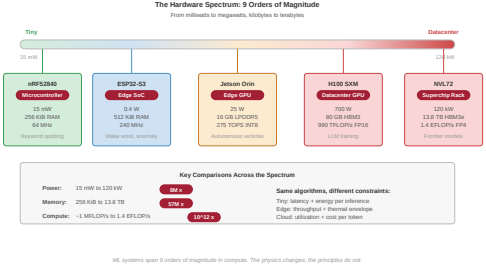
TinyML to Frontier

Key Question

Can the same analytical framework model a $2 microcontroller *and* a $3M GPU rack?

The answer is yes—because the **Roofline model** is universal.
Only the numbers change. The physics is the same.

The 9-Order-of-Magnitude Scale Span



Device	FLOPS	TDP
nRF52840	64 M	15 mW
ESP32-S3	500 M	400 mW
H100 SXM	989 T	700 W

Compute	$\sim 10^7\times$
Mem BW	$\sim 10^7\times$
Power	$\sim 10^{4.7}\times$

Same Roofline equation. Nine orders of magnitude apart.

Flash vs SRAM: The TinyML Memory Wall

Cloud GPU (H100)

- Weights in HBM (80 GB)
- Activations in SRAM (50 MB L2)
- Bandwidth: 3.35 TB/s
- Model $\leq$ 80 GB $\Rightarrow$ fits

TinyML MCU (ESP32-S3)

- Weights in **Flash** (8 MB)
- Activations in **SRAM** (512 KiB)
- Flash BW: 80 MB/s
- Model $\leq$ 8 MB $\Rightarrow$ fits
- Model $\leq$ 512 KiB $\Rightarrow$ SRAM-only (12$\times$ faster)

Key insight: mlsysim automatically selects the right memory tier:

1. Model fits in SRAM $\Rightarrow$ use SRAM bandwidth
2. Model in Flash $\Rightarrow$ use Flash bandwidth (bottleneck!)

Energy per Inference: μJ to Joules

Device	TDP	Energy/Inf	Use Case
nRF52840	15 mW	$\sim$50 μJ	Keyword spotting
ESP32-S3	400 mW	$\sim$1 mJ	Person detection
Jetson Orin NX	25 W	$\sim$25 mJ	Object detection
H100 SXM	700 W	$\sim$50 J	LLM inference

- 6 orders of magnitude in energy per inference
- TinyML: battery life is the primary constraint
- Duty cycling (sleep 99% of the time) extends battery to years

Live Demo: nRF52840 vs ESP32 vs H100

```
tiny_model = mlsysim.Models.Tiny.KeywordSpotting
devices = [mlsysim.Hardware.Tiny.nRF52840,
           mlsysim.Hardware.Tiny.ESP32_S3,
           mlsysim.Hardware.Cloud.H100]
for d in devices:
    p = mlsysim.Engine.solve(tiny_model, d,
          precision="int8", efficiency=0.3)
    print(f"{d.name:>15}: {p.latency.to('ms'):.2f~P}"
          f" | {p.bottleneck:<8}")
```

Observation

nRF52840 is memory-bound (Flash bottleneck). H100 finishes in μs—but wastes 700 W doing it. **Right-sizing hardware matters.** mlsysim also supports **Coral Edge TPU**,

Same Roofline, Different Physics

$$\text{Latency} = \max\left(\frac{\text{FLOPs}}{\text{Peak} \times \eta}, \frac{\text{Weights}}{\text{BW}}\right)$$

Term	Cloud	TinyML
Peak	989 TF	500 MF
BW	3.35 TB/s	80 MB/s
Overhead	0.01 ms	1.0 ms

```
# Same API, different hardware
for d in [mlsysim.Hardware.Cloud
    .H100,
          mlsysim.Hardware.Tiny.
    ESP32_S3]:
     p = mlsysim.Engine.solve(
         mlsysim.Models.Tiny
           .KeywordSpotting,
         d, precision="int8")
     print(f"{d.name}: {p.
    bottleneck}")
```

Both are memory-bound at batch size 1. Same equation, same diagnosis.

Exercise: Right-Sizing for IoT

Your Task (5 minutes)

A smart doorbell needs person detection at $<$100 ms latency on a coin-cell battery (500 mAh @ 3V = 1.5 Wh). Compare ESP32-S3 vs nRF52840: which device can run inference for a full year at 1 inference/minute?

```
import mlsysim

model = mlsysim.Models.Tiny.PersonDetection
for hw in [mlsysim.Hardware.Tiny.ESP32_S3,
           mlsysim.Hardware.Tiny.nRF52840]:
    p = mlsysim.Engine.solve(model, hw,
          precision="int8", efficiency=0.3)
    energy_j = p.energy.to("J").magnitude
    inferences_per_year = 60 * 24 * 365
```

Key Takeaway: TinyML to Frontier

1. The Roofline model spans **9 orders of magnitude**
2. **Right-size** hardware to the workload, not the other way around
3. At TinyML scale, **energy and memory** dominate; at cloud scale, **cost and communication**

"The best accelerator is the smallest one that meets your latency and accuracy SLA."

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Advanced Topics

4

Key Question

How do you compose multiple analytical models into an end-to-end system analysis?

Single-wall analysis answers “what is the bottleneck?”

Pipeline composition answers “what is the total cost of the whole stack?”

The Pipeline Composition Pattern

```python
from mlsysim.core.pipeline import Pipeline
from mlsysim.core.solver import (
    ScalingModel, DistributedModel, EconomicsModel)

# Chain: Algorithm -> Fleet -> Economics
pipe = Pipeline([
    ScalingModel(),       # Wall 11: compute budget
    DistributedModel(),   # Wall 14: comms overhead
    EconomicsModel(),     # Wall 17: total cost
])
print(pipe.explain())   # shows DAG + gaps
```

`explain()` shows:

Running the Pipeline

```
from mlsysim.core.constants import Q_

results = pipe.run(
    compute_budget=Q_("1e21 FLOP"),
    fleet=mlsysim.Systems.Clusters.Research_256,
    duration_days=30,
    datacenter=mlsysim.Infra.Grids.Quebec,
    mfu=0.45,
    infrastructure_multiplier=2.5)

for stage_name, result in results.items():
    print(f"\n--- {stage_name} ---")
    print(result)
```

Wall 21: Sensitivity Analysis

```
solver = mlsysim.SensitivitySolver()
result = solver.solve(
    model=llama, hardware=hw, precision="fp16",
    perturbation_pct=10.0, efficiency=0.5)
print(f"Binding: {result.binding_constraint}")
for p, s in result.sensitivities.items():
    tag = "<<<" if p == result.binding_constraint else ""
    print(f"  {p:>20}: {s:+.4f}  {tag}")
```

Rule: Invest in the parameter with the *largest* $|\partial T / \partial p|$.

Improving a non-binding parameter yields **zero** measurable gain.

Wall 22: Inverse Roofline (SynthesisSolver)

```python
from mlsysim.core.constants import Q_

solver = mlsysim.SynthesisSolver()
result = solver.solve(
    model=llama, target_latency=Q_("50 ms"),
    precision="fp16", efficiency=0.5)
print(f"Required BW:    {result.required_bw:.0f~P}")
print(f"Required FLOPS: {result.required_flops:.1f~P}")
```

Usage: “I need 50 ms TTFT. What hardware do I need?”

$\Rightarrow$ Derive specs from SLA, then match to real accelerators.

Fallacies & Pitfalls (Patterson Tradition)

Fallacy: "Cheaper hardware is always more cost-effective"

Reality: A 2× cheaper GPU that takes 3× longer has *higher* TCO.

TCO = CapEx + OpEx; slower hardware burns more electricity.

Fallacy: "More FLOPS = proportionally faster"

Reality: A100 → H100 is 6.3× peak FLOPS but only ∼2× for memory-bound LLM inference. The binding constraint is bandwidth, not compute.

Pitfall: Optimizing a non-binding parameter

If inference is memory-bound, doubling FLOPS gives 0% speedup.

Use `SensitivitySolver` to find the binding constraint *first*.

Pitfall: Ignoring the infrastructure multiplier

GPU cost is 40% of total datacenter TCO. Budgeting for GPUs alone

The Iron Law Revisited: All Five Terms

$$T = \frac{\text{FLOPs}}{N \times \text{Peak} \times \text{MFU} \times \eta_{\text{scale}} \times \text{Goodput}}$$

Term	What reduces it	Wall
N	Budget (buy more GPUs)	Wall 17
Peak	GPU generation (H100→B200)	Wall 1
MFU	FlashAttention, kernel fusion	Wall 3
η_{scale}	Network BW, gradient compression	Wall 14

Exercise: Binding Constraint Analysis

Your Task (5 minutes)

Use `SensitivitySolver` on Llama-3-8B + H100. Which parameter is binding?

Now switch to `batch_size=64` and re-run. Does the binding constraint change?

```
solver = mlsysim.SensitivitySolver()
# Batch size 1
r1 = solver.solve(llama, hw, efficiency=0.5)
print(f"bs=1:  binding = {r1.binding_constraint}")
# Batch size 64
p64 = mlsysim.Engine.solve(llama, hw, batch_size=64)
print(f"bs=64: bottleneck = {p64.bottleneck}")
```

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Wrap-up & Future

The 22 Walls of ML Systems

#	Wall	#	Wall
1	Compute	12	Reasoning (Emerging)
2	Memory	13	Fidelity (Compression)
3	Software (MFU)	14	Communication
4	Serving	15	Fragility (Reliability)
5	Batching (KV)	16	Multi-tenant (Queueing)
6	Streaming	17	Capital (TCO)
7	Tail Latency	18	Sustainability
8	Ingestion	19	Checkpoint
9	Transformation	20	Safety (Privacy)
10	Locality	21	Sensitivity
11	Complexity	22	Synthesis (Inverse)

What `mlsysim` Does *Not* Model

Not modeled (v0.1.0):

- Cache effects / tiling / fusion
- Real network congestion / jitter
- CUDA kernel scheduling
- Multi-model co-location
- Compiler optimizations

By design:

- Not a cycle-accurate simulator
- Not a profiler replacement
- Not a deployment tool
- **Is:** a first-principles analytical reasoning tool

Use it for:

- “Which constraint is binding?”
- “Is this hardware sufficient?”
- “What should I try first?”

v0.2.0 Roadmap

Models & Hardware

1. **MoE support** — Expert routing + parallelism
2. **GQA** / **MQA** — Grouped-query KV cache
3. **Network congestion** — Contention-aware collectives

Tooling & Integration

4. **Spot pricing** — Cloud cost optimization
5. **Browser exercises** — Live exploration
6. **HuggingFace Hub** — Auto-import models

Contributions welcome: `github.com/harvard-edge/mlsysim`

Design Challenge: Capstone Exercise

The Problem

$5M budget. Serve Llama-3 70B at **1,000 QPS** with $<$**100 ms TTFT** in **two regions** (US-East + EU-West). Design the fleet.

You must specify:

1. Hardware choice (which GPU? how many?)
2. Parallelism strategy (TP $\times$ PP)
3. Precision (FP16? INT8? FP8?)
4. Geographic placement (carbon?)
5. Redundancy (replicas per region?)

```python
import mlsysim

# Capstone starter code
model = mlsysim.Models.Language.
    Llama3_70B
hw = mlsysim.Hardware.H100

# Step 1: Can it fit?
p = mlsysim.Engine.solve(
    model, hw, precision="fp16")
```

Your Turn: Name That Wall

Think of one ML system in your own work.

Write down:

1. What is the system? (model + hardware + use case)
2. Which of the 22 walls is the **binding constraint**?
3. What would you change to move the wall?

2 minutes. Then we will hear from 3 volunteers.

Resources & Next Steps

Get Started

- `pip install mlsysim`
- GitHub: `harvard-edge/mlsysim`
- Website tutorials, examples, and API reference
- Full docs: `mlsysim.readthedocs.io`

The Textbook

- *Machine Learning Systems*
- Volume I: Foundations (single node)
- Volume II: Systems at Scale (fleet)

Key Papers

- Williams et al. (2009)
  Roofline Model
- Chowdhery et al. (2022)
  PaLM / MFU
- Hoffmann et al. (2022)
  Chinchilla Scaling
- Patterson et al. (2021)
  Carbon & Training
- Kwon et al. (2023)
  PagedAttention